Finally, let's talk about change data capture, a game-changer in the world of data management. Unlike traditional filesystems, CDC in Delta has been purposefully designed to handle data updates efficiently. Let's take a closer look at CDC and explore its capabilities through two practical scenarios:

- **Scenario 1 – effortless record updates**: Picture a scenario involving Rami, one of your customers. He initially made a purchase in Wisconsin but later relocated to Colorado, where he continued to make purchases. In your records, it's essential to reflect Rami's new address in Colorado. Here's where Delta's CDC shines. It effortlessly updates Rami's customer record without treating him as a new customer. CDC excels at capturing and applying updates seamlessly, ensuring data integrity without any hassles.
- **Scenario 2 – adapting to evolving data sources**: Now, consider a situation where your data source experiences unexpected changes, resulting in adding a new column containing information about your customers. Let's say this new column provides insights into the colors of items purchased by customers. This is valuable data that you wouldn't want to lose. Delta's CDC, combined with its schema evolution feature, comes to the rescue.

Schema evolution, explored in depth in *Chapter 3*, enables Delta to gracefully adapt to schema changes without causing any disruptions. When dealing with a new data column, Delta smoothly incorporates this information, ensuring your data remains up to date while retaining its full historical context. This ensures that you can leverage valuable insights for both present and future analyses.

Applying our learning

This book is heavily project-based. Each chapter starts with an overview of the important concepts and Data Intelligence Platform features that will prepare you for the main event – the *Applying our learning* sections. Every *Applying our learning* section has a *Technical requirements* section so that you know what technical resources you will need, in addition to your Databricks workspace and GitHub repository, to complete the project work in the respective chapter.

Technical requirements

Here are the technical requirements needed to get started with the hands-on examples used throughout this book:

- We use Kaggle for two of our datasets. If you do not already have an account, you will need to create one.
- Throughout the book, we will refer to code in GitHub. Create an account if you do not already have one.